ELSEWEDY UNIVERSITY OF TECHNOLOGY
- POLYTECHNIC OF EGYPT -

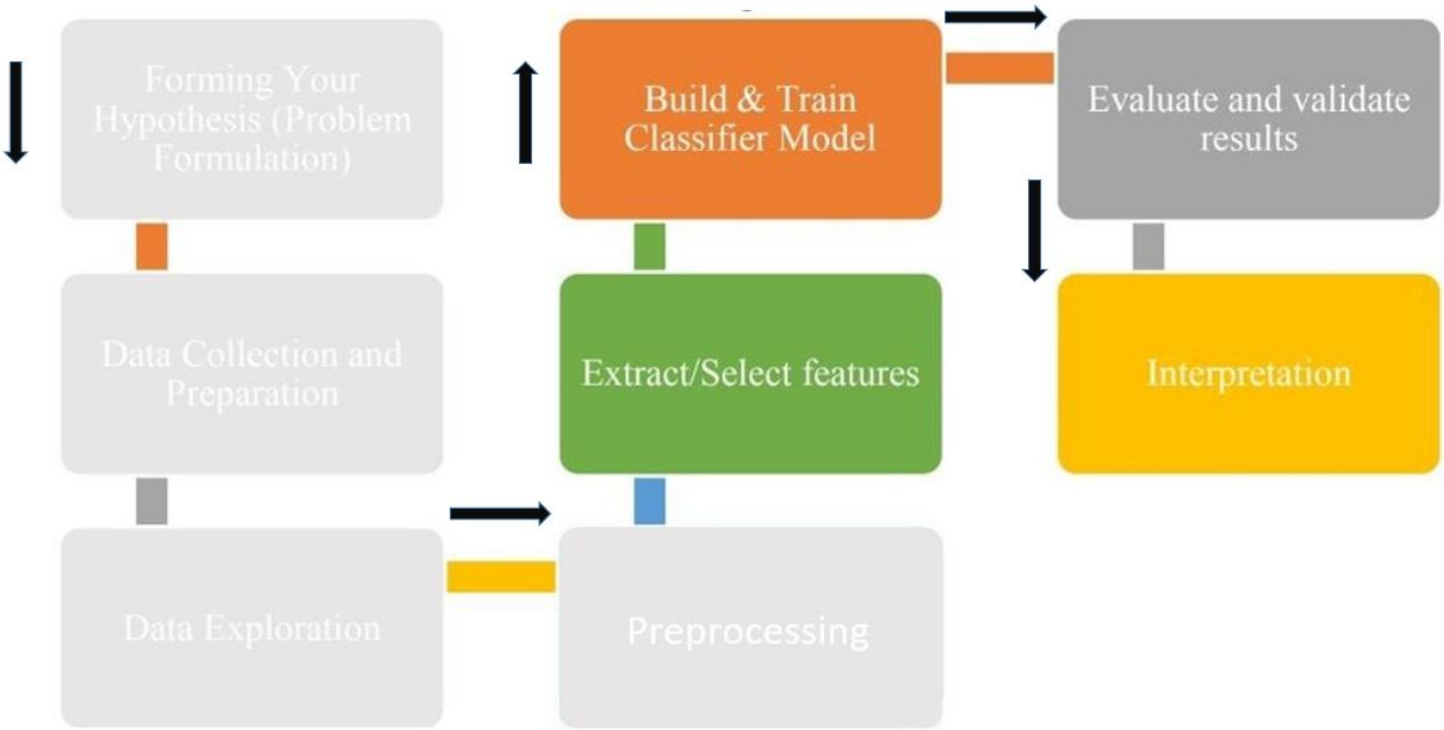
Algorithms for Data Science

Lecture 4: Algorithms for feature selection (Part 2)

© Spring 2026 – **Dr. Dalia Ezzat**
dalia.ezzat@sut.edu.eg

Table of Contents

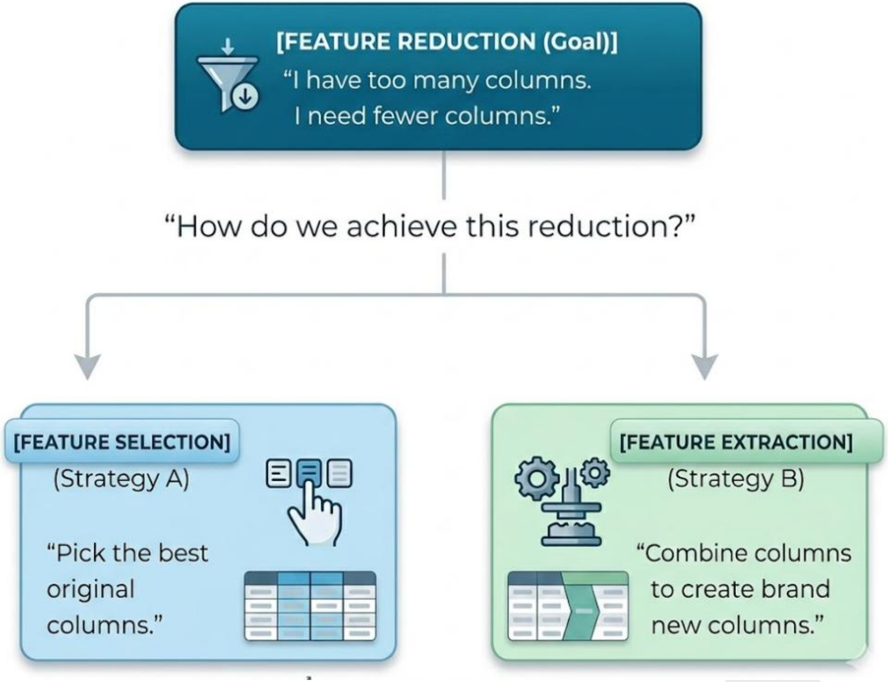
The Data Science Pipeline



Excerpts from the previous lecture (1/2)

- **Feature selection:** is the process of selecting a subset of the most relevant features from the original set of features while discarding the irrelevant or less important ones.

- **Common Feature Selection Techniques**
  - Filter Methods
  - Wrapper Methods
  - Embedded Methods



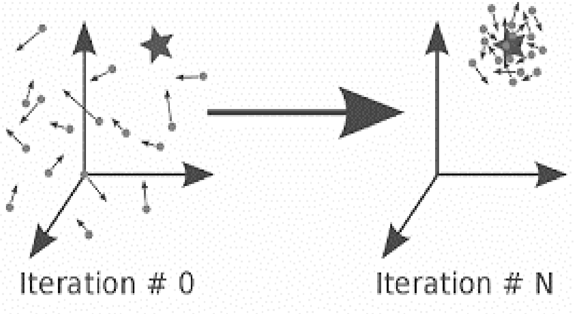
Excerpts from the previous lecture (2/2)

- **Two Types of Wrappers:**

Type	Approach	Algorithms
Deterministic	Follows a fixed, greedy path (add or remove features one by one).	Forward Selection, Backward Elimination, Recursive Feature Elimination (RFE)
Stochastic (Population-Based)	Uses a population of candidate solutions that explore the feature space collectively.	**Genetic Algorithms (GA)** , **Particle Swarm Optimization (PSO)**

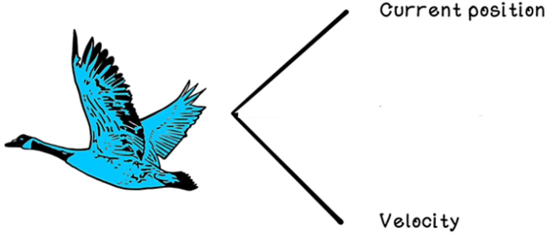
Particle Swarm Optimization

- A search technique used in computing to find optimal or near optimal solutions to optimization problems.
- PSO is an algorithm inspired from the nature social behavior and dynamic movements with communications of insects, birds and fish.
- Uses a number of particles that constitute a swarm moving around in the search space looking for the best solution.
- Each particle in search space adjusts its "flying" according to its own flying experience as well as the flying experience of other particles.



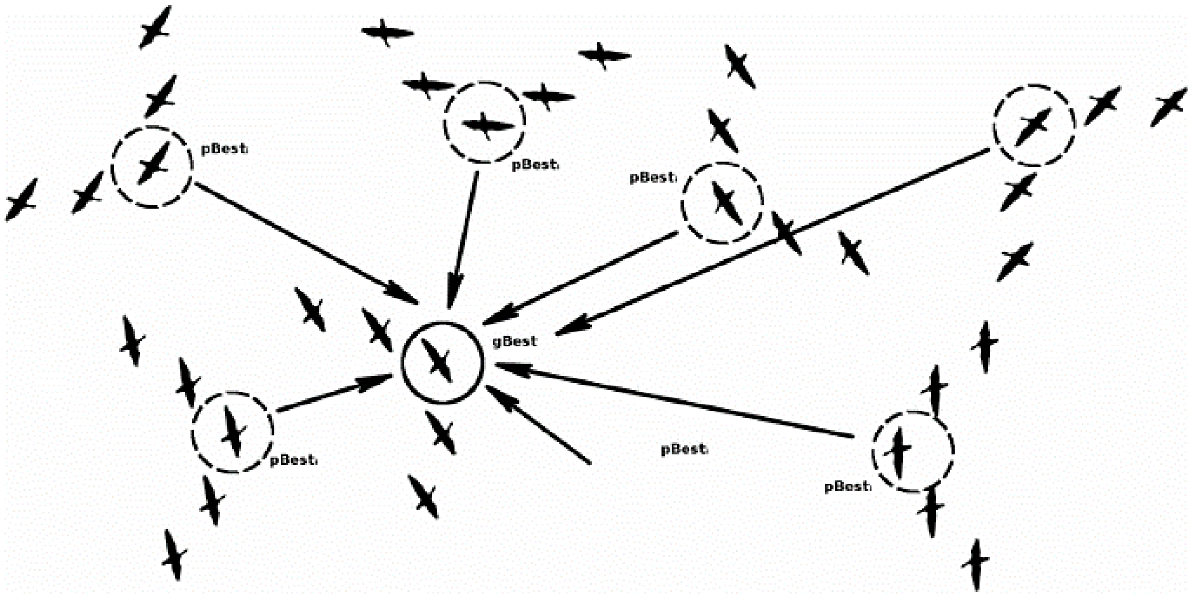
Particle Swarm Optimization- Terminologies

- **Population N Group of all particles.**
- **Fitness function (objective function): $f(x)$**
- **Particles i Any possible solution.**
- **Search Space [lb, ub] all possible solutions to the problem.**
- **Each particle is assumed to have two characteristics: A position X_i , A velocity V_i.**



Particle Swarm Optimization- Terminologies

- **Each particle keeps track: Personal best $Pbest_i$, and Global best $gbest$.**



PSO for Feature Selection

In the context of Feature Selection, each particle's position represents a subset of features.

Position Vector

$$\mathbf{x}_i = [x_i^1, x_i^2, \ldots, x_i^D]$$

where D is the total number of features, and $x_i^d \in \{0, 1\}$. $x_i^d = 1$ indicates feature d is selected, $x_i^d = 0$ indicates it is not selected.

Velocity Vector

$$\mathbf{v}_i = [v_i^1, v_i^2, \ldots, v_i^D]$$

where $v_i^d \in \mathbb{R}$. Velocity can be positive or negative:

- $v_i^d > 0$: tendency to select feature d ($S(v) > 0.5$)
- $v_i^d < 0$: tendency to exclude feature d ($S(v) < 0.5$)
- $v_i^d = 0$: equal probability ($S(0) = 0.5$)

PSO for Feature Selection

Sigmoid Function

$$S(v) = \frac{1}{1 + e^{-v}}$$

Velocity Update Equation

$$v_i^d(t+1) = w \cdot v_i^d(t) + c_1 r_1 \left(pbest_i^d - x_i^d(t)\right) + c_2 r_2 \left(gbest^d - x_i^d(t)\right)$$

Position Update Rule

$$x_i^d(t+1) = \begin{cases} 1 & \text{if } r < S\left(v_i^d(t+1)\right) \\ 0 & \text{otherwise} \end{cases}$$

PSO for Feature Selection

- **Fitness Function:** Many fitness function can be used either the problem is maximization or minimization, such as the following.

Type A (Maximization):

$$\text{Fitness}(\mathbf{x}_i) = \text{Accuracy}(\mathbf{x}_i)$$

Type B (Minimization with penalty):

$$\text{Fitness}(\mathbf{x}_i) = \alpha \cdot (1 - \text{Accuracy}(\mathbf{x}_i)) + (1 - \alpha) \cdot \frac{|\mathbf{x}_i|}{D}$$

where $|\mathbf{x}_i| = \sum_{d=1}^{D} x_i^d$ and $\alpha = 0.8$.

Steps of PSO Algorithm for Feature Selection

The PSO algorithm for feature selection proceeds through the following steps:

Step 1: Initialize Parameters

Set the algorithm parameters:

- w: inertia weight (typically between 0.4 and 1.2)
- c_1: cognitive acceleration coefficient
- c_2: social acceleration coefficient
- N: swarm size (number of particles)
- D: total number of features
- $T_{\max}$: maximum number of iterations
- $V_{\max}$: maximum velocity (typically 6)

Steps of PSO Algorithm for Feature Selection

Step 2: Initialize Swarm

For each particle $i = 1, 2, \dots, N$:

For each dimension $d = 1, 2, \dots, D$:

- Initialize position $x_i^d(1)$ randomly as either 0 or 1
- Initialize velocity $v_i^d(1)$ randomly in the range $[-V_{\max}, V_{\max}]$

Steps of PSO Algorithm for Feature Selection

Step 3: Evaluate Initial Fitness

For each particle $i = 1, 2, \ldots, N$:

- Extract the dataset using only the features where $x_i^d = 1$
- Train a classifier on the training data
- Evaluate on validation data to obtain Loss $(\mathbf{x}_i(1))$
- Compute fitness value $\text{Fitness}(\mathbf{x}_i(1))$
- Set personal best $\mathbf{pbest}_i(1) = \mathbf{x}_i(1)$

Steps of PSO Algorithm for Feature Selection

Step 4: Initialize Global Best

Find the particle with the best fitness value:

$$\mathbf{gbest}(1) = \arg\min_{\mathbf{pbest}_i} \text{Fitness}(\mathbf{pbest}_i(1))$$

Steps of PSO Algorithm for Feature Selection

Step 5: Iterate Until Termination

For each iteration $t = 1, 2, \ldots, T_{\text{max}}$:
For each particle $i = 1, 2, \ldots, N$:

Step 5.1: Generate Random Numbers

Generate three random numbers:

- $r_1 \sim U(0, 1)$ for cognitive component
- $r_2 \sim U(0, 1)$ for social component
- $r \sim U(0, 1)$ for position update threshold

Steps of PSO Algorithm for Feature Selection

Step 5.2: Update Velocity

For each dimension $d = 1, 2, \ldots, D$:

$$v_i^d(t+1) = w \cdot v_i^d(t) + c_1 r_1 \left(pbest_i^d - x_i^d(t)\right) + c_2 r_2 \left(gbest^d - x_i^d(t)\right)$$

If $v_i^d(t+1) > V_{\max}$, set $v_i^d(t+1) = V_{\max}$ If $v_i^d(t+1) < -V_{\max}$, set $v_i^d(t+1) = -V_{\max}$

Step 5.3: Apply Sigmoid Transformation

For each dimension $d = 1, 2, \ldots, D$:

$$S\left(v_i^d(t+1)\right) = \frac{1}{1 + e^{-v_i^d(t+1)}}$$

Steps of PSO Algorithm for Feature Selection

Step 5.4: Update Position

For each dimension $d = 1, 2, \ldots, D$:

$$x_i^d(t+1) = \begin{cases} 1 & \text{if } r < S\left(v_i^d(t+1)\right) \\ 0 & \text{otherwise} \end{cases}$$

Step 5.5: Evaluate New Position

- Extract the dataset using only the features where $x_i^d(t+1) = 1$
- Train a classifier on the training data
- Evaluate on validation data to obtain $\text{Loss}\left(\mathbf{x}_i(t+1)\right)$
- Compute fitness value $\text{Fitness}(\mathbf{x}_i(t+1))$

Steps of PSO Algorithm for Feature Selection

Step 5.6: Update Personal Best

If $\text{Fitness}(\mathbf{x}_i(t+1)) < \text{Fitness}(\mathbf{pbest}_i(t))$:

$$\mathbf{pbest}_i(t+1) = \mathbf{x}_i(t+1)$$

Otherwise:

$$\mathbf{pbest}_i(t+1) = \mathbf{pbest}_i(t)$$

Step 5.7: Update Global Best

Find the particle with the best fitness value among all personal bests:

$$\mathbf{gbest}(t+1) = \arg\min_{\mathbf{pbest}_i} \text{Fitness}(\mathbf{pbest}_i(t+1))$$

Steps of PSO Algorithm for Feature Selection

Step 6: Check Termination Criteria

Terminate if any of the following conditions are met:

- Maximum number of iterations $T_{\max}$ is reached
- Fitness value of **gbest** falls below a predefined threshold
- No improvement in **gbest** for a specified number of consecutive iterations

Step 7: Return Optimal Feature Subset

Return **gbest** as the optimal feature subset.

Let's take the first example

Example 1

Given Information

Consider a dataset with $D = 3$ features. A swarm of $N = 2$ particles is used for feature selection. The fitness function is:

$$\text{Fitness}(\mathbf{x}_i) = \text{Accuracy}(\mathbf{x}_i)$$

where **higher** values indicate better solutions.

Algorithm Parameters:

- Inertia weight: $w = 0.7$
- Cognitive coefficient: $c_1 = 1.5$
- Social coefficient: $c_2 = 1.5$

Example 1

Iteration 1 (Initialization) - Given Values

Particle 1:

- Initial position: $\mathbf{x}_1(1) = [1, 0, 1]$
- Initial velocity: $\mathbf{v}_1(1) = [0.2, -0.1, 0.4]$
- Classification accuracy: $\text{Accuracy}(\mathbf{x}_1(1)) = 0.85$

Particle 2:

- Initial position: $\mathbf{x}_2(1) = [0, 1, 0]$
- Initial velocity: $\mathbf{v}_2(1) = [-0.3, 0.5, -0.2]$
- Classification accuracy: $\text{Accuracy}(\mathbf{x}_2(1)) = 0.75$

Example 1

Iteration 2 Update - Given Values

For Particle 1 Update:

- Random number for cognitive component: $r_1 = 0.6$
- Random number for social component: $r_2 = 0.2$
- Random number for position threshold: $r = 0.5$
- If Particle 1 moves to a new position, its accuracy will be: $\text{Accuracy}(\mathbf{x}_1(2)) = 0.85$

Example 1

Iteration 2 Update - Given Values

For Particle 2 Update:

- Random number for cognitive component: $r_1 = 0.4$
- Random number for social component: $r_2 = 0.9$
- Random number for position threshold: $r = 0.3$
- If Particle 2 moves to a new position, its accuracy will be: $\text{Accuracy}(\mathbf{x}_2(2)) = 0.82$

Example 1

Additional given values for calculations:

- $e^{-0.14} = 0.8694$
- $e^{0.07} = 1.0725$
- $e^{-0.28} = 0.7558$
- $e^{-1.14} = 0.3198$
- $e^{1.00} = 2.7183$
- $e^{-1.21} = 0.2982$

Example 1

Required Calculations

Perform two complete iterations of the PSO algorithm:

Iteration 1 Tasks:

1. Calculate the fitness values for both particles at Iteration 1.
2. Determine the personal best $\mathbf{pbest}_1(1)$ and $\mathbf{pbest}_2(1)$.
3. Determine the global best $\mathbf{gbest}(1)$.

Example 1

Iteration 2 Tasks for Particle 1:

4. Calculate the updated velocity $\mathbf{v}_1(2)$ for Particle 1.
5. Calculate the sigmoid transformed values $S(v_1^d(2))$ for each dimension.
6. Determine the updated position $\mathbf{x}_1(2)$ for Particle 1.
7. Calculate the fitness value Fitness$(\mathbf{x}_1(2))$.
8. Determine if the personal best $\mathbf{pbest}_1(2)$ is updated.

Example 1

Iteration 2 Tasks for Particle 2:

9. Calculate the updated velocity $\mathbf{v}_2(2)$ for Particle 2.
10. Calculate the sigmoid transformed values $S(v_2^d(2))$ for each dimension.
11. Determine the updated position $\mathbf{x}_2(2)$ for Particle 2.
12. Calculate the fitness value $\text{Fitness}(\mathbf{x}_2(2))$.
13. Determine if the personal best $\mathbf{pbest}_2(2)$ is updated.

Final Tasks:

14. Determine the global best $\mathbf{gbest}(2)$ at the end of Iteration 2.

Example 1

Solution

Iteration 1 Results

Task 1: Fitness Values at Iteration 1

For Particle 1:

$$\text{Fitness}(\mathbf{x}_1(1)) = \text{Accuracy}(\mathbf{x}_1(1)) = 0.85$$

For Particle 2:

$$\text{Fitness}(\mathbf{x}_2(1)) = \text{Accuracy}(\mathbf{x}_2(1)) = 0.75$$

Task 2: Personal Best at Iteration 1

Since this is the first iteration, each particle's personal best is its initial position:

$$\mathbf{pbest}_1(1) = \mathbf{x}_1(1) = [1, 0, 1], \quad \text{Fitness} = 0.85$$

$$\mathbf{pbest}_2(1) = \mathbf{x}_2(1) = [0, 1, 0], \quad \text{Fitness} = 0.75$$

Task 3: Global Best at Iteration 1

Comparing fitness values (higher is better):

$$\text{Fitness}(\mathbf{pbest}_1(1)) = 0.85 > 0.75 = \text{Fitness}(\mathbf{pbest}_2(1))$$

Therefore:

$$\mathbf{gbest}(1) = \mathbf{pbest}_1(1) = [1, 0, 1]$$

Example 1

Iteration 2: Update Particle 1

Task 4: Velocity Update for Particle 1

For dimension $d = 1$:

$$v_1^1(2) = w \cdot v_1^1(1) + c_1 r_1 \left(pbest_1^1 - x_1^1(1)\right) + c_2 r_2 \left(gbest^1 - x_1^1(1)\right)$$

Substituting $w = 0.7$, $v_1^1(1) = 0.2$, $c_1 = 1.5$, $r_1 = 0.6$, $pbest_1^1 = 1$, $x_1^1(1) = 1$, $c_2 = 1.5$, $r_2 = 0.2$, $gbest^1 = 1$:

$$\begin{aligned} v_1^1(2) &= 0.7 \times 0.2 + 1.5 \times 0.6 \times (1 - 1) + 1.5 \times 0.2 \times (1 - 1) \\ &= 0.14 + 1.5 \times 0.6 \times 0 + 1.5 \times 0.2 \times 0 \\ &= 0.14 + 0 + 0 = 0.14 \end{aligned}$$

Example 1

For dimension $d = 2$:

$$v_1^2(2) = w \cdot v_1^2(1) + c_1 r_1 \left(pbest_1^2 - x_1^2(1)\right) + c_2 r_2 \left(gbest^2 - x_1^2(1)\right)$$

Substituting $v_1^2(1) = -0.1$, $pbest_1^2 = 0$, $x_1^2(1) = 0$, $gbest^2 = 0$:

$$\begin{aligned} v_1^2(2) &= 0.7 \times (-0.1) + 1.5 \times 0.6 \times (0 - 0) + 1.5 \times 0.2 \times (0 - 0) \\ &= -0.07 + 1.5 \times 0.6 \times 0 + 1.5 \times 0.2 \times 0 \\ &= -0.07 + 0 + 0 = -0.07 \end{aligned}$$

Example 1

For dimension $d = 3$:

$$v_1^3(2) = w \cdot v_1^3(1) + c_1 r_1 \left(pbest_1^3 - x_1^3(1)\right) + c_2 r_2 \left(gbest^3 - x_1^3(1)\right)$$

Substituting $v_1^3(1) = 0.4$, $pbest_1^3 = 1$, $x_1^3(1) = 1$, $gbest^3 = 1$:

$$\begin{aligned} v_1^3(2) &= 0.7 \times 0.4 + 1.5 \times 0.6 \times (1 - 1) + 1.5 \times 0.2 \times (1 - 1) \\ &= 0.28 + 1.5 \times 0.6 \times 0 + 1.5 \times 0.2 \times 0 \\ &= 0.28 + 0 + 0 = 0.28 \end{aligned}$$

The updated velocity vector is:

$$\mathbf{v}_1(2) = [0.14, -0.07, 0.28]$$

Example 1

Task 5: Sigmoid Transformation for Particle 1

For dimension $d = 1$ with $v = 0.14$:

$$S(0.14) = \frac{1}{1 + e^{-0.14}} = \frac{1}{1 + 0.8694} = \frac{1}{1.8694} = 0.535$$

For dimension $d = 2$ with $v = -0.07$:

$$S(-0.07) = \frac{1}{1 + e^{0.07}} = \frac{1}{1 + 1.0725} = \frac{1}{2.0725} = 0.483$$

For dimension $d = 3$ with $v = 0.28$:

$$S(0.28) = \frac{1}{1 + e^{-0.28}} = \frac{1}{1 + 0.7558} = \frac{1}{1.7558} = 0.569$$

Example 1

Task 6: Position Update for Particle 1

For dimension $d = 1$, compare $r = 0.5$ with $S(0.14) = 0.535$: Since $0.5 < 0.535$, set $x_1^1(2) = 1$

For dimension $d = 2$, compare $r = 0.5$ with $S(-0.07) = 0.483$: Since $0.5 > 0.483$, set $x_1^2(2) = 0$

For dimension $d = 3$, compare $r = 0.5$ with $S(0.28) = 0.569$: Since $0.5 < 0.569$, set $x_1^3(2) = 1$

The updated position vector is:

$$\mathbf{x}_1(2) = [1, 0, 1]$$

Example 1

Task 7: Fitness for Particle 1 at Iteration 2

Given $\text{Accuracy}(\mathbf{x}_1(2)) = 0.85$:

$$\text{Fitness}(\mathbf{x}_1(2)) = 0.85$$

Task 8: Personal Best Update for Particle 1

Compare with current personal best fitness of 0.85:

$$\text{Fitness}(\mathbf{x}_1(2)) = 0.85 \quad \text{vs} \quad \text{Fitness}(\mathbf{pbest}_1(1)) = 0.85$$

Since 0.85 is not less than 0.85, the personal best does not change:

$$\mathbf{pbest}_1(2) = \mathbf{pbest}_1(1) = [1, 0, 1]$$

Example 1

Iteration 2: Update Particle 2

Task 9: Velocity Update for Particle 2

For dimension $d = 1$:

$$v_2^1(2) = w \cdot v_2^1(1) + c_1 r_1 \left(pbest_2^1 - x_2^1(1)\right) + c_2 r_2 \left(gbest^1 - x_2^1(1)\right)$$

Substituting $v_2^1(1) = -0.3$, $pbest_2^1 = 0$, $x_2^1(1) = 0$, $gbest^1 = 1$, $r_1 = 0.4$, $r_2 = 0.9$:

$$\begin{aligned} v_2^1(2) &= 0.7 \times (-0.3) + 1.5 \times 0.4 \times (0 - 0) + 1.5 \times 0.9 \times (1 - 0) \\ &= -0.21 + 1.5 \times 0.4 \times 0 + 1.5 \times 0.9 \times 1 \\ &= -0.21 + 0 + 1.35 = 1.14 \end{aligned}$$

Example 1

For dimension $d = 2$:

$$v_2^2(2) = w \cdot v_2^2(1) + c_1 r_1 \left(pbest_2^2 - x_2^2(1)\right) + c_2 r_2 \left(gbest^2 - x_2^2(1)\right)$$

Substituting $v_2^2(1) = 0.5$, $pbest_2^2 = 1$, $x_2^2(1) = 1$, $gbest^2 = 0$:

$$\begin{aligned} v_2^2(2) &= 0.7 \times 0.5 + 1.5 \times 0.4 \times (1 - 1) + 1.5 \times 0.9 \times (0 - 1) \\ &= 0.35 + 1.5 \times 0.4 \times 0 + 1.5 \times 0.9 \times (-1) \\ &= 0.35 + 0 - 1.35 = -1.00 \end{aligned}$$

For dimension $d = 3$:

$$v_2^3(2) = w \cdot v_2^3(1) + c_1 r_1 \left(pbest_2^3 - x_2^3(1)\right) + c_2 r_2 \left(gbest^3 - x_2^3(1)\right)$$

Substituting $v_2^3(1) = -0.2$, $pbest_2^3 = 0$, $x_2^3(1) = 0$, $gbest^3 = 1$:

$$\begin{aligned} v_2^3(2) &= 0.7 \times (-0.2) + 1.5 \times 0.4 \times (0 - 0) + 1.5 \times 0.9 \times (1 - 0) \\ &= -0.14 + 1.5 \times 0.4 \times 0 + 1.5 \times 0.9 \times 1 \\ &= -0.14 + 0 + 1.35 = 1.21 \end{aligned}$$

The updated velocity vector is:

$$\mathbf{v}_2(2) = [1.14, -1.00, 1.21]$$

Example 1

Task 10: Sigmoid Transformation for Particle 2

For dimension $d = 1$ with $v = 1.14$:

$$S(1.14) = \frac{1}{1 + e^{-1.14}} = \frac{1}{1 + 0.3198} = \frac{1}{1.3198} = 0.758$$

For dimension $d = 2$ with $v = -1.00$:

$$S(-1.00) = \frac{1}{1 + e^{1.00}} = \frac{1}{1 + 2.7183} = \frac{1}{3.7183} = 0.269$$

For dimension $d = 3$ with $v = 1.21$:

$$S(1.21) = \frac{1}{1 + e^{-1.21}} = \frac{1}{1 + 0.2982} = \frac{1}{1.2982} = 0.770$$

Example 1

Task 11: Position Update for Particle 2

For dimension $d = 1$, compare $r = 0.3$ with $S(1.14) = 0.758$: Since $0.3 < 0.758$, set $x_2^1(2) = 1$

For dimension $d = 2$, compare $r = 0.3$ with $S(-1.00) = 0.269$: Since $0.3 > 0.269$, set $x_2^2(2) = 0$

For dimension $d = 3$, compare $r = 0.3$ with $S(1.21) = 0.770$: Since $0.3 < 0.770$, set $x_2^3(2) = 1$

The updated position vector is:

$$\mathbf{x}_2(2) = [1, 0, 1]$$

Example 1

Task 12: Fitness for Particle 2 at Iteration 2

Given $\text{Accuracy}(\mathbf{x}_2(2)) = 0.82$:

$$\text{Fitness}(\mathbf{x}_2(2)) = 0.82$$

Task 13: Personal Best Update for Particle 2

Compare with current personal best fitness of 0.75:

$$\text{Fitness}(\mathbf{x}_2(2)) = 0.82 \quad \text{vs} \quad \text{Fitness}(\mathbf{pbest}_2(1)) = 0.75$$

Since $0.82 > 0.75$ (better), the personal best IS updated:

$$\mathbf{pbest}_2(2) = \mathbf{x}_2(2) = [1, 0, 1]$$

$$\text{Fitness}(\mathbf{pbest}_2(2)) = 0.82$$

Example 1

Global Best at Iteration 2

Task 14: Global Best Update

Compare all personal bests at Iteration 2:

$$\text{Fitness}(\mathbf{pbest}_1(2)) = 0.85$$

$$\text{Fitness}(\mathbf{pbest}_2(2)) = 0.82$$

Since $0.85 > 0.82$, Particle 1 still has the better fitness:

$$\mathbf{gbest}(2) = \mathbf{pbest}_1(2) = [1, 0, 1]$$

Let's take the second example

Example 2

Consider a dataset with $D = 3$ features. A swarm of $N = 2$ particles is used for feature selection. The fitness function balances accuracy and feature count:

$$\text{Fitness}(\mathbf{x}_i) = 0.8 \times (1 - \text{Accuracy}(\mathbf{x}_i)) + 0.2 \times \frac{|\mathbf{x}_i|}{3}$$

where $|\mathbf{x}_i| = \sum_{d=1}^{3} x_i^d$. For this formulation, a **lower** fitness value indicates a better solution.

Algorithm Parameters:

- Inertia weight: $w = 0.7$
- Cognitive coefficient: $c_1 = 1.5$
- Social coefficient: $c_2 = 1.5$
- Trade-off weight: $\alpha = 0.8$

Example 2

Iteration 1 (Initialization) - Given Values

Particle 1:

- Initial position: $\mathbf{x}_1(1) = [1, 0, 1]$
- Initial velocity: $\mathbf{v}_1(1) = [0.2, -0.1, 0.4]$
- Classification accuracy: $\text{Accuracy}(\mathbf{x}_1(1)) = 0.85$

Particle 2:

- Initial position: $\mathbf{x}_2(1) = [0, 1, 0]$
- Initial velocity: $\mathbf{v}_2(1) = [-0.3, 0.5, -0.2]$
- Classification accuracy: $\text{Accuracy}(\mathbf{x}_2(1)) = 0.75$

Example 2

Iteration 2 Update - Given Values

For Particle 1 Update:

- Random number for cognitive component: $r_1 = 0.6$
- Random number for social component: $r_2 = 0.2$
- Random number for position threshold: $r = 0.5$
- If Particle 1 moves to a new position, its accuracy will be: $\text{Accuracy}(\mathbf{x}_1(2)) = 0.85$

Example 2

For Particle 2 Update:

- Random number for cognitive component: $r_1 = 0.4$
- Random number for social component: $r_2 = 0.9$
- Random number for position threshold: $r = 0.3$
- If Particle 2 moves to a new position, its accuracy will be: $\text{Accuracy}(\mathbf{x}_2(2)) = 0.82$

Additional given values for calculations:

- $e^{-0.14} = 0.8694$
- $e^{0.07} = 1.0725$
- $e^{-0.28} = 0.7558$
- $e^{-1.14} = 0.3198$
- $e^{1.00} = 2.7183$
- $e^{-1.21} = 0.2982$

Example 2

Required Calculations

Perform two complete iterations of the PSO algorithm:

Iteration 1 Tasks:

1. Calculate the number of selected features $|\mathbf{x}_1(1)|$ and $|\mathbf{x}_2(1)|$.
2. Calculate the fitness values for both particles at Iteration 1.
3. Determine the personal best $\mathbf{pbest}_1(1)$ and $\mathbf{pbest}_2(1)$.
4. Determine the global best $\mathbf{gbest}(1)$.

Example 2

Iteration 2 Tasks for Particle 1:

5. Calculate the updated velocity $\mathbf{v}_1(2)$ for Particle 1.

6. Calculate the sigmoid transformed values $S(v_1^d(2))$ for each dimension.

7. Determine the updated position $\mathbf{x}_1(2)$ for Particle 1.

8. Calculate the number of selected features $|\mathbf{x}_1(2)|$.

9. Calculate the fitness value $\text{Fitness}(\mathbf{x}_1(2))$.

10. Determine if the personal best $\mathbf{pbest}_1(2)$ is updated.

Example 2

Iteration 2 Tasks for Particle 2:

11. Calculate the updated velocity $\mathbf{v}_2(2)$ for Particle 2.
12. Calculate the sigmoid transformed values $S(v_2^d(2))$ for each dimension.
13. Determine the updated position $\mathbf{x}_2(2)$ for Particle 2.
14. Calculate the number of selected features $|\mathbf{x}_2(2)|$.
15. Calculate the fitness value Fitness$(\mathbf{x}_2(2))$.
16. Determine if the personal best $\mathbf{pbest}_2(2)$ is updated.

Final Tasks:

17. Determine the global best $\mathbf{gbest}(2)$ at the end of Iteration 2.

Example 2

Solution

Iteration 1 Results

Task 1: Number of Selected Features at Iteration 1

For Particle 1:

$$|\mathbf{x}_1(1)| = x_1^1 + x_1^2 + x_1^3 = 1 + 0 + 1 = 2$$

For Particle 2:

$$|\mathbf{x}_2(1)| = x_2^1 + x_2^2 + x_2^3 = 0 + 1 + 0 = 1$$

Task 2: Fitness Values at Iteration 1

For Particle 1:

$$\begin{aligned}
\text{Fitness}(\mathbf{x}_1(1)) &= 0.8 \times (1 - 0.85) + 0.2 \times \frac{2}{3} \\
&= 0.8 \times 0.15 + 0.2 \times 0.666667 \\
&= 0.12 + 0.133333 = 0.253333
\end{aligned}$$

For Particle 2:

$$\begin{aligned}
\text{Fitness}(\mathbf{x}_2(1)) &= 0.8 \times (1 - 0.75) + 0.2 \times \frac{1}{3} \\
&= 0.8 \times 0.25 + 0.2 \times 0.333333 \\
&= 0.2 + 0.066667 = 0.266667
\end{aligned}$$

Example 2

Task 3: Personal Best at Iteration 1

Since this is the first iteration, each particle's personal best is its initial position:

$$\mathbf{pbest}_1(1) = \mathbf{x}_1(1) = [1, 0, 1], \quad \text{Fitness} = 0.253333$$

$$\mathbf{pbest}_2(1) = \mathbf{x}_2(1) = [0, 1, 0], \quad \text{Fitness} = 0.266667$$

Task 4: Global Best at Iteration 1

Comparing fitness values (lower is better):

$$\text{Fitness}(\mathbf{pbest}_1(1)) = 0.253333 < 0.266667 = \text{Fitness}(\mathbf{pbest}_2(1))$$

Therefore:

$$\mathbf{gbest}(1) = \mathbf{pbest}_1(1) = [1, 0, 1]$$

Example 2

Iteration 2: Update Particle 1

Task 5: Velocity Update for Particle 1

For dimension $d = 1$:

$$v_1^1(2) = w \cdot v_1^1(1) + c_1 r_1 \left(pbest_1^1 - x_1^1(1)\right) + c_2 r_2 \left(gbest^1 - x_1^1(1)\right)$$

Substituting $w = 0.7$, $v_1^1(1) = 0.2$, $c_1 = 1.5$, $r_1 = 0.6$, $pbest_1^1 = 1$, $x_1^1(1) = 1$, $c_2 = 1.5$, $r_2 = 0.2$, $gbest^1 = 1$:

$$\begin{aligned} v_1^1(2) &= 0.7 \times 0.2 + 1.5 \times 0.6 \times (1 - 1) + 1.5 \times 0.2 \times (1 - 1) \\ &= 0.14 + 1.5 \times 0.6 \times 0 + 1.5 \times 0.2 \times 0 \\ &= 0.14 + 0 + 0 = 0.14 \end{aligned}$$

Example 2

For dimension $d = 2$:

$$v_1^2(2) = w \cdot v_1^2(1) + c_1 r_1 \left(pbest_1^2 - x_1^2(1)\right) + c_2 r_2 \left(gbest^2 - x_1^2(1)\right)$$

Substituting $v_1^2(1) = -0.1$, $pbest_1^2 = 0$, $x_1^2(1) = 0$, $gbest^2 = 0$:

$$\begin{aligned} v_1^2(2) &= 0.7 \times (-0.1) + 1.5 \times 0.6 \times (0 - 0) + 1.5 \times 0.2 \times (0 - 0) \\ &= -0.07 + 1.5 \times 0.6 \times 0 + 1.5 \times 0.2 \times 0 \\ &= -0.07 + 0 + 0 = -0.07 \end{aligned}$$

For dimension $d = 3$:

$$v_1^3(2) = w \cdot v_1^3(1) + c_1 r_1 \left(pbest_1^3 - x_1^3(1)\right) + c_2 r_2 \left(gbest^3 - x_1^3(1)\right)$$

Substituting $v_1^3(1) = 0.4$, $pbest_1^3 = 1$, $x_1^3(1) = 1$, $gbest^3 = 1$:

$$\begin{aligned} v_1^3(2) &= 0.7 \times 0.4 + 1.5 \times 0.6 \times (1 - 1) + 1.5 \times 0.2 \times (1 - 1) \\ &= 0.28 + 1.5 \times 0.6 \times 0 + 1.5 \times 0.2 \times 0 \\ &= 0.28 + 0 + 0 = 0.28 \end{aligned}$$

The updated velocity vector is:

$$\mathbf{v}_1(2) = [0.14, -0.07, 0.28]$$

Example 2

Task 6: Sigmoid Transformation for Particle 1

For dimension $d = 1$ with $v = 0.14$:

$$S(0.14) = \frac{1}{1 + e^{-0.14}} = \frac{1}{1 + 0.8694} = \frac{1}{1.8694} = 0.535$$

For dimension $d = 2$ with $v = -0.07$:

$$S(-0.07) = \frac{1}{1 + e^{0.07}} = \frac{1}{1 + 1.0725} = \frac{1}{2.0725} = 0.483$$

For dimension $d = 3$ with $v = 0.28$:

$$S(0.28) = \frac{1}{1 + e^{-0.28}} = \frac{1}{1 + 0.7558} = \frac{1}{1.7558} = 0.569$$

Example 2

Task 7: Position Update for Particle 1

For dimension $d = 1$, compare $r = 0.5$ with $S(0.14) = 0.535$: Since $0.5 < 0.535$, set $x_1^1(2) = 1$

For dimension $d = 2$, compare $r = 0.5$ with $S(-0.07) = 0.483$: Since $0.5 > 0.483$, set $x_1^2(2) = 0$

For dimension $d = 3$, compare $r = 0.5$ with $S(0.28) = 0.569$: Since $0.5 < 0.569$, set $x_1^3(2) = 1$

The updated position vector is:

$$\mathbf{x}_1(2) = [1, 0, 1]$$

Example 2

Task 8: Number of Selected Features for Particle 1 at Iteration 2

$$|\mathbf{x}_1(2)| = 1 + 0 + 1 = 2$$

Task 9: Fitness for Particle 1 at Iteration 2

Given Accuracy$(\mathbf{x}_1(2)) = 0.85$:

$$\begin{aligned}
\text{Fitness}(\mathbf{x}_1(2)) &= 0.8 \times (1 - 0.85) + 0.2 \times \frac{2}{3} \\
&= 0.8 \times 0.15 + 0.2 \times 0.666667 \\
&= 0.12 + 0.133333 = 0.253333
\end{aligned}$$

Task 10: Personal Best Update for Particle 1

Compare with current personal best fitness of 0.253333:

$$\text{Fitness}(\mathbf{x}_1(2)) = 0.253333 \quad \text{vs} \quad \text{Fitness}(\mathbf{pbest}_1(1)) = 0.253333$$

Since they are equal, the personal best does not change:

$$\mathbf{pbest}_1(2) = \mathbf{pbest}_1(1) = [1, 0, 1]$$

Example 2

Iteration 2: Update Particle 2

Task 11: Velocity Update for Particle 2

For dimension $d = 1$:

$$v_2^1(2) = w \cdot v_2^1(1) + c_1 r_1 \left(pbest_2^1 - x_2^1(1)\right) + c_2 r_2 \left(gbest^1 - x_2^1(1)\right)$$

Substituting $v_2^1(1) = -0.3$, $pbest_2^1 = 0$, $x_2^1(1) = 0$, $gbest^1 = 1$, $r_1 = 0.4$, $r_2 = 0.9$:

$$\begin{aligned} v_2^1(2) &= 0.7 \times (-0.3) + 1.5 \times 0.4 \times (0 - 0) + 1.5 \times 0.9 \times (1 - 0) \\ &= -0.21 + 1.5 \times 0.4 \times 0 + 1.5 \times 0.9 \times 1 \\ &= -0.21 + 0 + 1.35 = 1.14 \end{aligned}$$

For dimension $d = 2$:

$$v_2^2(2) = w \cdot v_2^2(1) + c_1 r_1 \left(pbest_2^2 - x_2^2(1)\right) + c_2 r_2 \left(gbest^2 - x_2^2(1)\right)$$

Substituting $v_2^2(1) = 0.5$, $pbest_2^2 = 1$, $x_2^2(1) = 1$, $gbest^2 = 0$:

$$\begin{aligned} v_2^2(2) &= 0.7 \times 0.5 + 1.5 \times 0.4 \times (1 - 1) + 1.5 \times 0.9 \times (0 - 1) \\ &= 0.35 + 1.5 \times 0.4 \times 0 + 1.5 \times 0.9 \times (-1) \\ &= 0.35 + 0 - 1.35 = -1.00 \end{aligned}$$

Example 2

For dimension $d = 3$:

$$v_2^3(2) = w \cdot v_2^3(1) + c_1 r_1 \left(pbest_2^3 - x_2^3(1)\right) + c_2 r_2 \left(gbest^3 - x_2^3(1)\right)$$

Substituting $v_2^3(1) = -0.2$, $pbest_2^3 = 0$, $x_2^3(1) = 0$, $gbest^3 = 1$:

$$\begin{aligned} v_2^3(2) &= 0.7 \times (-0.2) + 1.5 \times 0.4 \times (0 - 0) + 1.5 \times 0.9 \times (1 - 0) \\ &= -0.14 + 1.5 \times 0.4 \times 0 + 1.5 \times 0.9 \times 1 \\ &= -0.14 + 0 + 1.35 = 1.21 \end{aligned}$$

The updated velocity vector is:

$$\mathbf{v}_2(2) = [1.14, -1.00, 1.21]$$

Example 2

Task 12: Sigmoid Transformation for Particle 2

For dimension $d = 1$ with $v = 1.14$:

$$S(1.14) = \frac{1}{1 + e^{-1.14}} = \frac{1}{1 + 0.3198} = \frac{1}{1.3198} = 0.758$$

For dimension $d = 2$ with $v = -1.00$:

$$S(-1.00) = \frac{1}{1 + e^{1.00}} = \frac{1}{1 + 2.7183} = \frac{1}{3.7183} = 0.269$$

For dimension $d = 3$ with $v = 1.21$:

$$S(1.21) = \frac{1}{1 + e^{-1.21}} = \frac{1}{1 + 0.2982} = \frac{1}{1.2982} = 0.770$$

Example 2

Task 13: Position Update for Particle 2

For dimension $d = 1$, compare $r = 0.3$ with $S(1.14) = 0.758$: Since $0.3 < 0.758$, set $x_2^1(2) = 1$

For dimension $d = 2$, compare $r = 0.3$ with $S(-1.00) = 0.269$: Since $0.3 > 0.269$, set $x_2^2(2) = 0$

For dimension $d = 3$, compare $r = 0.3$ with $S(1.21) = 0.770$: Since $0.3 < 0.770$, set $x_2^3(2) = 1$

The updated position vector is:

$$\mathbf{x}_2(2) = [1, 0, 1]$$

Task 14: Number of Selected Features for Particle 2 at Iteration 2

$$|\mathbf{x}_2(2)| = 1 + 0 + 1 = 2$$

Task 15: Fitness for Particle 2 at Iteration 2

Given $\text{Accuracy}(\mathbf{x}_2(2)) = 0.82$:

$$\begin{aligned}
\text{Fitness}(\mathbf{x}_2(2)) &= 0.8 \times (1 - 0.82) + 0.2 \times \frac{2}{3} \\
&= 0.8 \times 0.18 + 0.2 \times 0.666667 \\
&= 0.144 + 0.133333 = 0.277333
\end{aligned}$$

Example 2

Task 16: Personal Best Update for Particle 2

Compare with current personal best fitness of 0.266667:

$$\text{Fitness}(\mathbf{x}_2(2)) = 0.277333 \quad \text{vs} \quad \text{Fitness}(\mathbf{pbest}_2(1)) = 0.266667$$

Since $0.277333 > 0.266667$ (worse), the personal best does NOT change:

$$\mathbf{pbest}_2(2) = \mathbf{pbest}_2(1) = [0, 1, 0]$$

Global Best at Iteration 2

Task 17: Global Best Update

Compare all personal bests at Iteration 2:

$$\text{Fitness}(\mathbf{pbest}_1(2)) = 0.253333$$

$$\text{Fitness}(\mathbf{pbest}_2(2)) = 0.266667$$

Since $0.253333 < 0.266667$, Particle 1 still has the better fitness:

$$\mathbf{gbest}(2) = \mathbf{pbest}_1(2) = [1, 0, 1]$$

Any Question?